

Forecasting Daily Satellite Sea-Surface-Temperature Images of the Brazil–Malvinas Confluence with PINNeAPLe Neural Operators

A leakage-controlled study with 43 years of NOAA OISST v2.1 and VeriPhysics-style validation

PINNeAPLe Labs

PINNeAPLe Labs — Physics AI research notes

September 25, 2026 · Experiment 01

Abstract

We train neural-operator and convolutional forecasters from the PINNeAPLe library to reproduce, day by day, satellite images of sea-surface temperature (SST) one to seven days ahead. The data are 15,706 daily NOAA OISST v2.1 fields (AVHRR-based, 0.25°) over the Brazil–Malvinas Confluence ($48\text{--}32^\circ\text{S}$, $64\text{--}48^\circ\text{W}$), 1982–2024. Leakage is controlled by a strictly chronological split (train 1982–2012, validation 2013–2017, test 2018–2024), a climatology and normalisation fitted on training years only, model selection on validation only and a single final evaluation on test. Against the strongest trivial baselines, the validation-selected three-seed ResNet-CNN ensemble (PINNeAPLe Conv2DModel) reduces the test RMSE by 11.6% at day +1 (0.291 vs 0.329 $^\circ\text{C}$ for persistence) and by 12.5% at day +7 (0.929 vs 1.062 $^\circ\text{C}$; 8.7% vs damped persistence), with non-overlapping 95% block-bootstrap intervals. A Fourier Neural Operator (PINNeAPLe FNO2d) is also skilful but consistently second. The skill over persistence is nearly identical in the training, validation and test periods, which is the operational signature of absent overfitting. Split-conformal intervals calibrated on validation reach 91.8% empirical coverage at the nominal 90% on test. The main limitation is spectral: the forecasts lose power at intermediate scales (eddies), i.e. they are smoother than the satellite images. Validation metrics are aggregated with the VeriPhysics confidence score; checks that do not apply to a data-driven image forecaster are reported as not run.

Keywords: sea surface temperature; satellite imagery; neural operators; FNO; forecasting; overfitting control; conformal prediction; PINNeAPLe; VeriPhysics

What this study shows — and what it does not

- Shows: PINNeAPLe models beat persistence, damped persistence and climatology at every lead (1–7 d) on a 7-year held-out test period.
- Shows: the gain is reproducible across seeds and consistent across training, validation and test periods (no sign of overfitting).
- Does not show: skill beyond 7 days, skill in other ocean regions, or physical consistency of the forecasts — no physical law is imposed, and the predicted fields are measurably smoother than the observed ones.
- OISST is a Level-4 analysis (satellite + in situ, optimally interpolated), not raw swath imagery; the models learn to forecast that analysis.

1 Introduction

Daily SST maps derived from satellites are a standard input to ocean monitoring, fisheries and weather forecasting. The Brazil–Malvinas Confluence, where the warm Brazil Current meets the cold Malvinas (Falklands) Current, is one of the most energetic regions of the world ocean: fronts meander and shed eddies on time scales of days, which makes short-range forecasting non-trivial and persistence a surprisingly hard baseline to beat [1].

The goal here is not a new architecture but a trustworthy measurement: how much better than trivial forecasts can general-purpose PINNeAPLe neural operators reproduce the next satellite images, when every source of information leakage is closed and every number is tied to an executed check.

2 Data

We use NOAA Optimum Interpolation SST version 2.1 (OISST) [2], a daily 0.25° Level-4 product that blends AVHRR infrared satellite retrievals with ship and buoy observations. A 64×64 subset (lat -47.875 to -32.125° , lon -63.875 to -48.125°) was downloaded through the NOAA PSL THREDDS/OPeNDAP server for 1982-01-01 to 2024-12-31 (15,706 consecutive days, no gaps). 76.5% of the pixels are ocean; land pixels (South America) are masked everywhere. Figure 1 shows one field and its anomaly.

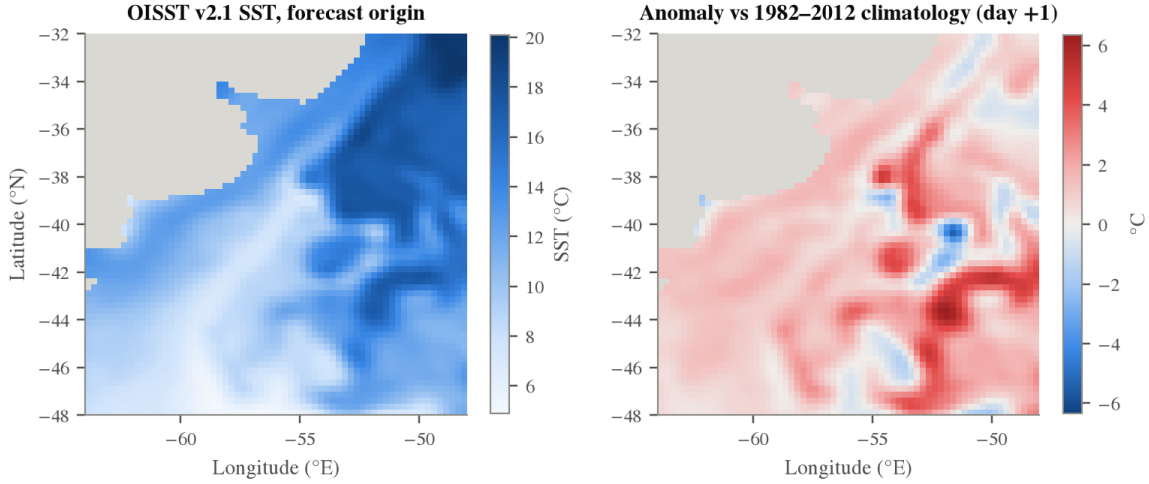


Figure 1. Study region. Left: one daily OISST v2.1 field at a forecast origin in July 2021 (test period). Right: its anomaly one day later relative to the harmonic climatology fitted on 1982–2012. Grey: land (masked).

2.1 Pre-processing and leakage control

- **Climatology:** per pixel, a mean plus three annual harmonics fitted by least squares on the training years only; anomalies = SST – climatology.
- **Normalisation:** anomalies divided by one scalar, their training-period standard deviation (1.167°C).
- **Samples:** input = 5 consecutive daily anomaly maps + land mask + sin/cos of day of year; target = the next 7 daily maps. A sample is kept only if its 12 days lie inside one split: 11,312 training, 1,815 validation and 2,546 test forecast origins.
- **Splits** are chronological blocks (no random shuffling of days), so autocorrelation cannot leak targets into training.

3 Methods

3.1 Models

All learned models predict the increment of the anomaly relative to the last observed day (residual form), so the zero function reproduces persistence. The loss is the mean squared error over ocean pixels and the 7 leads.

- **FNO2d** (PINNeAPLe *pinneapple_neural.architectures.neural_operators.FNO2d*): Fourier layers with grid coordinates as extra channels.
- **ResNet-CNN** (PINNeAPLe *Conv2DModel*): residual convolutional blocks with group normalisation.
- **Baselines:** persistence (anomaly held constant), climatology (zero anomaly) and damped persistence — a per-pixel, per-lead least-squares AR(1) coefficient fitted on training years.

3.2 Training, model selection and ensembling

Adam with decoupled weight decay (10^{-4}), cosine learning-rate schedule, gradient clipping, batch 32, early stopping on validation loss. Two configurations per family were compared on validation (Table 1); the best of each family was retrained with three seeds and the seed average is the ensemble forecast. The model reported as *selected* was chosen on validation RMSE before the test set was scored.

Table 1. Hyper-parameter comparison on the validation period (normalised masked MSE averaged over leads 1–7).

Family	Configuration	Parameters	Epochs	Best val. MSE
FNO	width=32, modes=12, layers=4	1,184,455	5	0.5016
FNO	width=48, modes=20, layers=4	7,383,079	5	0.4971
CNN	hidden=48, blocks=6	251,335	14	0.4827
CNN	hidden=64, blocks=8, dropout=0.1	593,927	21	0.4719

3.3 Metrics

RMSE and MAE (°C) over ocean pixels; anomaly correlation coefficient (ACC, spatial, per forecast); mean bias; radially averaged power-spectrum ratio of forecast to observation (a blurring diagnostic); 95% confidence intervals from a monthly block bootstrap (1000 resamples) that respects temporal autocorrelation.

3.4 Uncertainty and VeriPhysics validation

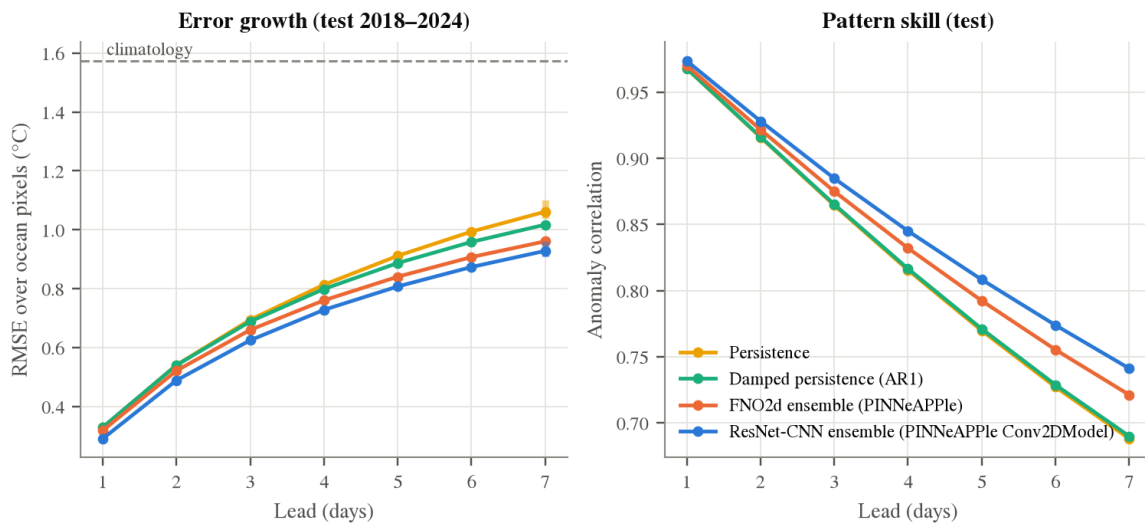
Two uncertainty estimates are calibrated on validation and evaluated on test: (i) per-lead split-conformal intervals using PINNeAPPLE's *ConformalPredictor* on per-pixel residuals, and (ii) per-pixel Gaussian σ estimated from validation residuals, scored with PINNeAPPLE's *CalibrationMetrics* (expected calibration error, ECE, and 90% coverage). The held-out test observations are registered as a named benchmark in the PINNeAPPLE physics database and compared through *PhysicsCase.validate_against_benchmark()*; the results are aggregated by *compute_physics_confidence* (VeriPhysics trust score), which reports the fraction of possible checks that actually ran.

4 Results

4.1 Forecast skill on the held-out period

Table 2. Test-period (2018–2024) scores, °C. ACC: anomaly correlation. Bias = forecast – observation.

Model	RMSE d+1	RMSE d+3	RMSE d+7	95% CI d+7	ACC d+7	Bias d+7
Persistence	0.329	0.695	1.062	1.049–1.089	0.688	1.26e-04
Climatology	1.573	1.574	1.574	1.536–1.635	0.000	-0.737
Damped persistence (AR1)	0.330	0.689	1.018	1.006–1.040	0.690	-0.261
FNO2d ensemble (PINNeAPPLE)	0.319	0.660	0.961	0.952–0.981	0.721	-0.178
ResNet-CNN ensemble (PINNeAPPLE Conv2DModel) (selected)	0.291	0.625	0.929	0.920–0.949	0.741	-0.139

**Figure 2.** Test RMSE and anomaly correlation versus lead. Vertical bars at day 7: 95% block-bootstrap intervals for the selected model and persistence.

The selected ResNet-CNN ensemble is the best model at every lead. Its advantage over persistence is 11.6% at day +1 and 12.5% at day +7; the bootstrap intervals do not overlap at either lead. Damped persistence captures part of the gain at longer leads (it learns that anomalies decay), but the learned models add spatial information — advection and front displacement — that a per-pixel statistic cannot. The FNO2d ensemble is skilful (0.961 °C at day +7) but trails the CNN at all leads, as it did on validation.

The climatology baseline has a mean bias of -0.737 °C in the test period: SST in 2018–2024 is warmer than the 1982–2012 climatology. Because all learned models are conditioned on the last observed anomaly, this shift does not propagate into their forecasts (bias -0.139 °C at day +7 for the selected model).

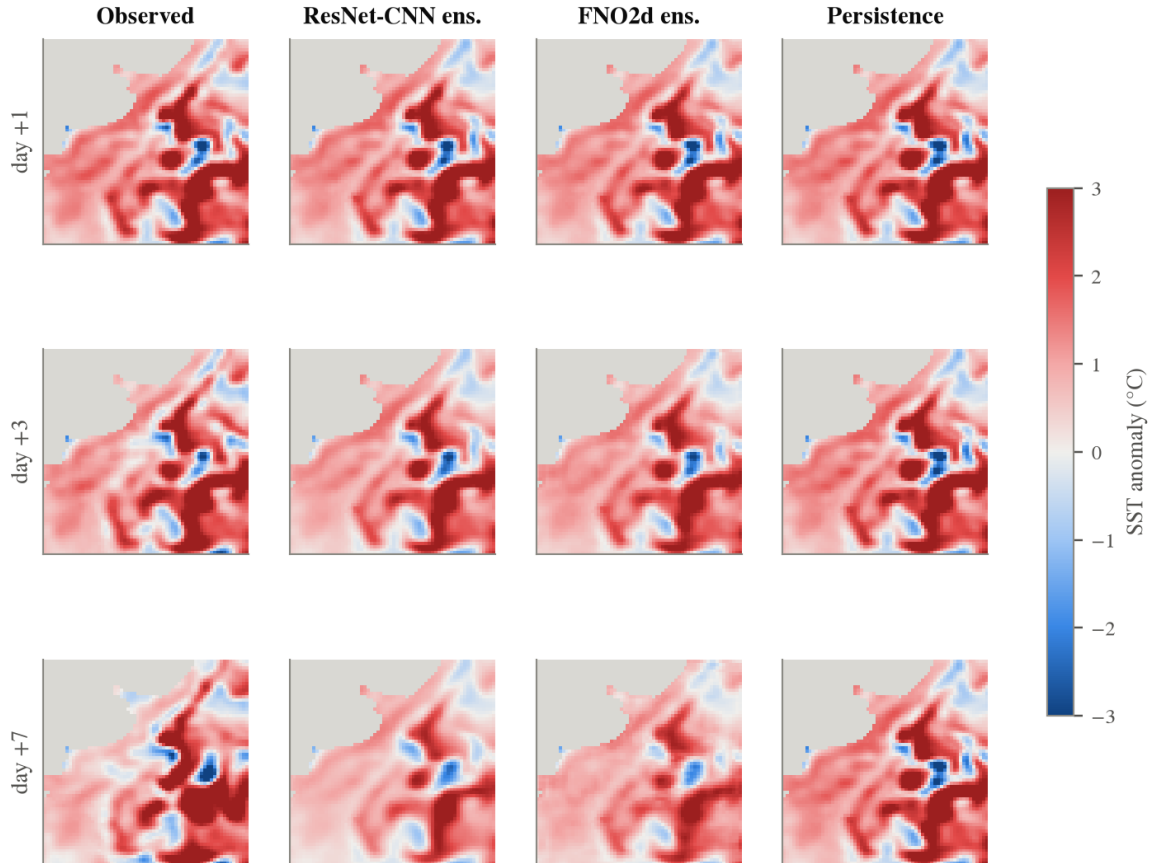


Figure 3. One test forecast (origin July 2021). Rows: leads 1, 3, 7 days. Columns: observed anomaly, the two PINNeAPLe ensembles and persistence. Colour scale clipped at ± 3 °C.

4.2 Overfitting diagnostics

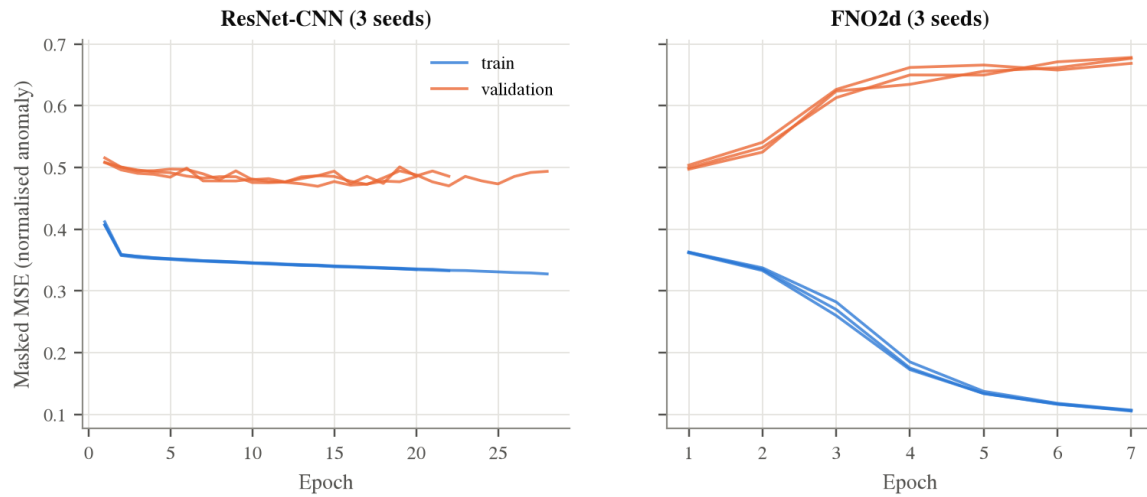


Figure 4. Training and validation loss of the final models (three seeds each). Validation loss plateaus while training loss keeps decreasing; early stopping keeps the checkpoint with the lowest validation loss.

Absolute errors differ between periods because the ocean itself is more or less predictable in each period (persistence RMSE at day +7 is 0.980, 1.160 and 1.062 °C in train, validation and test). The relevant quantity is therefore the skill relative to persistence, which is 14.1%, 13.6% and 12.5% respectively (Figure 5). A model that memorised the training period would show a large train–test gap in this ratio; none is present. Across seeds the test RMSE at day +7 of the three CNN members spans 0.930–0.937 °C.

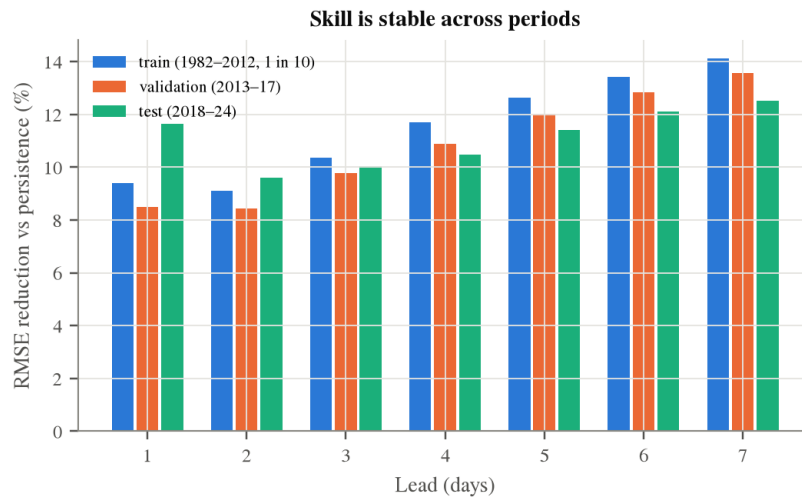


Figure 5. RMSE reduction of the selected model relative to persistence in each period. The training bars use one origin in ten.

Table 3. Test RMSE at day +7 by season of the forecast origin (°C). The ranking of models is the same in every season.

Model	DJF	MAM	JJA	SON
Persistence	1.066	1.027	1.095	1.087
Damped persistence (AR1)	1.000	1.005	1.050	1.035
FNO2d ensemble (PINNeAPLe)	0.961	0.941	0.983	0.980
ResNet-CNN ensemble (PINNeAPLe Conv2DModel)	0.924	0.915	0.958	0.940

4.3 Spectral fidelity

Figure 6 compares the power spectrum of the day-7 forecasts with that of the observations. All learned forecasts retain less than half of the observed power at intermediate wavenumbers: they predict where the anomalies will be but smooth the eddy-scale structure. This is the expected behaviour of a mean-squared-error forecaster under uncertainty (it predicts the conditional mean) and is the main reason these forecasts should not be read as realistic individual images at long leads.

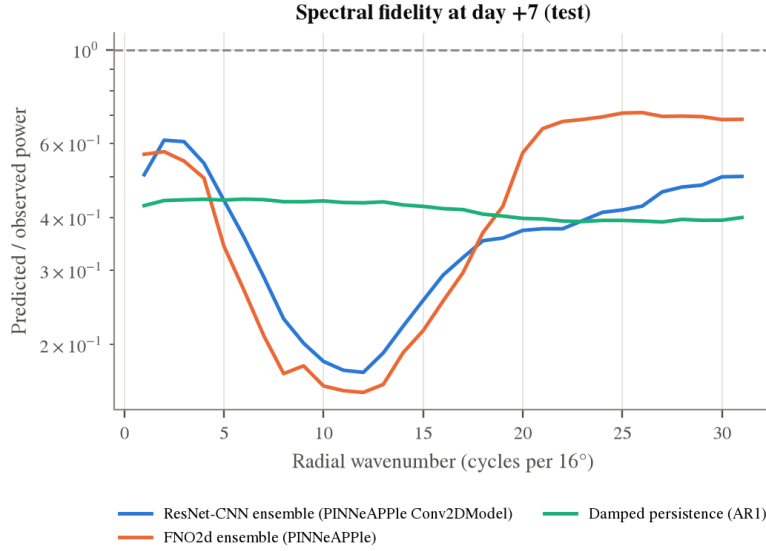


Figure 6. Ratio of forecast to observed radially averaged power at day +7 (1 = perfect spectral fidelity).

4.4 Uncertainty calibration

Table 4. Uncertainty calibrated on validation, evaluated on test, selected model.

Lead	Conformal half-width (°C)	Conformal coverage (target 0.90)	Gaussian $\bar{\sigma}$ (°C)	Gaussian coverage 90%	ECE
d+1	0.573	0.946	0.368	0.954	0.083
d+3	1.151	0.932	0.701	0.936	0.047
d+7	1.611	0.918	0.957	0.918	0.049

Both interval types slightly over-cover on test (they are conservative), consistent with the validation period being harder than the test period for every model, persistence included.

4.5 VeriPhysics validation summary

Table 5. VeriPhysics confidence (PINNeAPLe *compute_physics_confidence*). Scores must be read with their coverage (fraction of the five possible checks that ran). Not run: physics guardrail (no governing equation is imposed), numerical convergence (no discretisation parameter), geometry OOD (no geometry input).

Model	Trust score	Coverage	Relative L2 vs held-out obs.	Components
Persistence	0.519	0.2	0.481	benchmark_agreement
Climatology	0.000	0.2	1.000	benchmark_agreement
Damped persistence (AR1)	0.533	0.2	0.467	benchmark_agreement
FNO2d ensemble (PINNeAPLe)	0.556	0.2	0.444	benchmark_agreement
ResNet-CNN ensemble (PINNeAPLe Conv2DModel)	0.763	0.4	0.425	uq_calibration, benchmark_agreement

5 Discussion and limitations

- The improvement over persistence ($\approx 12\%$ RMSE) is modest in absolute terms, as expected for a strongly persistent field over 1–7 days; it is, however, robust to seeds, seasons and periods.

- Forecasts are smoother than observations (Section 4.3). Probabilistic or generative training would be required to produce realistic individual images; it was out of scope.
- Only one region and one product were tested. OISST's optimal interpolation already smooths small scales, so the task is easier than forecasting raw Level-2 imagery.
- Compute: models were trained on an Apple-silicon laptop shared with other jobs; wall-clock times in the logs reflect that contention, not the models.
- No physics loss was used. The comparison between purely data-driven and physics-informed variants is left for work where a governing equation (e.g. a mixed-layer heat budget with reanalysis currents) can be specified honestly.

6 Reproducibility

Code: `PINNeAPLe-Climate/experiments/01_sst_satellite_forecasting/src` (download_oisst.py, sst_experiment.py); every number in this report is read from results/results.json by paper/make_paper.py. Random seeds are fixed; test data were scored once.

References

- [1] Olson, D. B., Podestá, G. P., Evans, R. H., Brown, O. B. (1988). Temporal variations in the separation of Brazil and Malvinas Currents. *Deep-Sea Research* 35(12), 1971–1990.
- [2] Huang, B., et al. (2021). Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) Version 2.1. *Journal of Climate* 34(8), 2923–2939.
- [3] Li, Z., et al. (2021). Fourier Neural Operator for parametric partial differential equations. *ICLR*.
- [4] He, K., et al. (2016). Deep residual learning for image recognition. *CVPR*.
- [5] Vovk, V., Gammerman, A., Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer (conformal prediction).
- [6] NOAA PSL OISST v2.1 data access: <https://psl.noaa.gov/data/gridded/data.noaa.oisst.v2.highres.html>
- [7] PINNeAPLe: <https://github.com/PINNeAPLe-Labs/PINNeAPLe>